

7	(a)(i)	Photoelectric effect is the ejection/emission of an electron from a <u>metal</u> surface when the surface is irradiated with electromagnetic radiation of a high enough frequency.	A1																		
	(a)(ii)	Allow 350 nm to 400 nm	A1																		
	(a)(iii)	Threshold frequency is determined when there is no responsivity from photocathode/ no release of photoelectrons. From Fig. 7.2, threshold wavelength is approximately 640 nm. Threshold frequency is found by using speed of light divided by threshold wavelength (or 630 nm)	B1 B1																		
	(a)(iv)	Using $hc/\lambda = \text{work function} + E_{k, \text{max}}$, $hc/(450 \times 10^{-9}) = h(4.76 \times 10^{14}) + 0.5(9.11 \times 10^{-31})v^2$ $v = 5.27 \times 10^5 \text{ m s}^{-1}$ *	C1 C1 A1																		
	(a)(v)	Using a low work function photocathode will allow electrons to be emitted easily with photons of larger wavelength.	A1																		
	(b)(i)	Current = nq/t $7.8 \times 10^{-4} = (n/t)(1.6 \times 10^{-19})$ $n/t = 4.88 \times 10^{15} \text{ s}^{-1}$	C1 A1																		
	(b)(ii)	Number of photons per unit time $= 4.88 \times 10^{15} / (0.01)$ $= 4.88 \times 10^{17}$	A1																		
	(b)(iii)	4.8 mA of current is detected from photocathode when 1 W of power of EM radiation is incident on it.	A1																		
	(c)(i)	Use wavelength = 225 nm, $R = 9.0 \text{ mA/W}$ and wavelength = 540 nm (or 550 nm), $R = 22 \text{ mA/W}$ to correctly determine η at 5 %.	B1 B1																		
	(c)(ii)	Correct computation of values with maximum values of 2sf To deduct one mark for one wrong calculation. <table border="1"><thead><tr><th>η / %</th><th>λ / nm</th><th>R / mA W⁻¹</th></tr></thead><tbody><tr><td>3</td><td>300</td><td>7.2</td></tr><tr><td>3</td><td>400</td><td>9.7</td></tr><tr><td>3</td><td>500</td><td>12</td></tr><tr><td>3</td><td>600</td><td>14</td></tr><tr><td>3</td><td>700</td><td>17</td></tr></tbody></table>	η / %	λ / nm	R / mA W ⁻¹	3	300	7.2	3	400	9.7	3	500	12	3	600	14	3	700	17	A1 A1
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	(c)(iii)	Correct plotted points with best-fit curve	B1 B1																		
	(d)(i)	The voltage used in PMT is higher than 1000V and must be electrically insulated for safety reason. OR Any reasonable answer	B1																		
	(d)(ii)	- Use in eye devices to measure interruption of light. OR - To detect radiation. OR - Measure intensity and spectrum of light emitting materials in laboratory. https://en.wikipedia.org/wiki/Photomultiplier_tube#Usage_considerations	B1																		